TasteRank Explorer

Summary

Plain-Language Overview

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Companion documents: Technical Appendix · Methods Primer · Data Appendix · Grape Reference

Abstract. *TasteRank is a network-analytic framework for ranking wine grape varieties by structural centrality in sensory space. Each of 101 grape varieties is encoded as a 13-dimensional sensory profile vector; pairwise cosine similarities define a sparse k-nearest-neighbour graph with* $k = 5$ *(101 nodes, 341 edges); the eigenvector centrality of each node in this weighted graph is its TasteRank score. The procedure identifies which varieties sit at the dense centre of the tasting universe, surrounded by many similar peers (high TasteRank), and which sit at the periphery with distinctive, less-shared profiles (low TasteRank). Sagrantino, Nero d'Avola, and Lagrein occupy the densest core; Riesling, Sauvignon Blanc, and other noble varieties rank near the bottom precisely because their distinctive profiles are not widely shared. Six communities are detected by greedy modularity optimisation (*$Q \approx 0.41$*); the spectral gap* $|\lambda_2/\lambda_1| \approx 0.72$ *confirms a robust core–periphery structure. PageRank applied as a robustness check produces a Spearman rank correlation of* $\rho \approx 0.92$ *with TasteRank, with informative divergences identifying bridge varieties between communities.*

1. Introduction

Wine grape varieties are traditionally classified by colour (red vs. white), region of origin, or stylistic convention. While these categories are useful heuristics, they obscure a deeper structural reality: grape varieties exist in a continuous space of sensory similarity, where some varieties occupy central, archetypal positions and others sit at the periphery.

TasteRank is a network-analytic framework that makes this structure explicit. The framework rests on a simple act of translation. Each grape variety is described by a sensory profile — thirteen dimensions that have, in one form or another, organised structured wine tasting since Peynaud's work at Bordeaux and the analytical traditions that grew up around UC Davis: colour depth, aromatic intensity, the balance of fruit and herb, the architecture of acid, tannin, body, and finish. The choice of these particular thirteen, and the scores assigned to each variety, are the author's own; the underlying vocabulary is the common inheritance of modern oenology. Once a variety becomes a vector in this thirteen-dimensional space, the rest follows as geometry. Cosine similarity measures how alike two profiles are. A sparse graph of nearest neighbours makes the resulting structure tractable. Eigenvector centrality, applied to that graph, ranks varieties not by reputation or rarity but by how densely they sit at the centre of the tasting universe — how many other varieties resemble them, and how central those resemblances are in turn.

The result is a principled, reproducible ranking that answers a question no wine textbook addresses directly: which grape varieties are the most typical — that is, which varieties sit at the dense centre of sensory space, surrounded by many other similar varieties? High TasteRank identifies the archetypes; low TasteRank identifies the outliers.

2. Data

2.1 Grape Variety Universe

The dataset comprises 101 grape varieties (53 red, 48 white) selected for global significance, regional diversity, and representation across the canonical literature of wine education. The selection spans the canonical international varieties (Cabernet Sauvignon, Chardonnay, Pinot Noir) through to specialised regional varieties (Xinomavro, Assyrtiko, Blaufränkisch, Plavac Mali). The full list of varieties with profile scores is provided in the companion document *Data Appendix* (Section D.3); per-variety descriptive notes are in the *Grape Reference*.

2.2 Sensory Profile Encoding

Each variety is encoded as a thirteen-dimensional vector. The dimensions are drawn from the structured tasting tradition that runs from Peynaud through the modern descriptive frameworks (UC Davis, AWRI, and the various national wine schools that grew from them); they were chosen to span the sensory axes that experienced tasters consistently report as discriminating between varieties, while remaining few enough to be scored reliably on a coarse 0–5 scale. The thirteen dimensions used here are:

#	Dimension	Description
1	Colour Depth	Intensity of pigmentation (pale to opaque)
2	Aromatic Intensity	Strength of nose from low to pronounced
3	Floral Character	Degree of floral aromatic contribution
4	Fruit Ripeness	Ripe/dried fruit vs. underripe/green fruit
5	Herbal/Earthy	Green herb, earth, mineral notes
6	Spice/Oak	Oak-derived and spice character
7	Acidity	Perceived acidity from low to high
8	Tannin	Tannin level (reds); 0 for most whites
9	Body	Weight and texture on palate
10	Alcohol	Perceived alcohol warmth
11	Flavour Intensity	Palate flavour concentration
12	Finish	Length and persistence of aftertaste
13	Complexity	Layeredness and evolution of character

Scores reflect canonical varietal typicity — the profile a well-made, representative example of the variety would be expected to present — and are the author's own judgments, calibrated against the standard ampelographic literature (Robinson, Harding & Vouillamoz; d'Agata; Galet) and against years of structured tasting. They describe the variety, not any particular wine or vintage. A given Burgundian Pinot Noir might score higher in earth and lower in fruit ripeness than the canonical profile suggests; an Oregon Pinot might invert that relationship. The TasteRank score is best read as a statement about the variety's centre of gravity, not its full range.

3. Methodology

3.1 Cosine Similarity

Pairwise similarity between grape varieties is measured using cosine similarity on the thirteen-dimensional sensory profile vectors. For varieties i and j with profile vectors $\mathbf{p}_i$ and $\mathbf{p}_j$:

$$\cos(\mathbf{p}_i, \mathbf{p}_j) = \frac{\mathbf{p}_i \cdot \mathbf{p}_j}{\|\mathbf{p}_i\| \, \|\mathbf{p}_j\|}.$$

Cosine similarity is preferred over Euclidean distance because it measures angular proximity in profile space, making it invariant to the overall magnitude of scores and sensitive only to the shape of the sensory profile. Two varieties with proportionally similar profiles will have high cosine similarity even if one consistently scores higher across all dimensions.

3.2 k-Nearest-Neighbour Graph Construction

From the full 101×101 similarity matrix, a sparse k-nearest-neighbour graph with $k = 5$ is constructed. For each variety, the five most similar varieties are connected by edges weighted by their cosine similarity. The graph is symmetrised using maximum-weight union: if variety A includes B in its top-5 or B includes A in its top-5, the edge exists with weight equal to their cosine similarity. This produces a graph with 101 nodes and 341 edges — substantially sparser than the complete graph (which would have $\sim$5,050 edges) but rich enough to reveal community structure and centrality patterns.

3.3 Eigenvector Centrality (TasteRank)

TasteRank is defined as the eigenvector centrality of each node in the weighted kNN graph. For the weighted adjacency matrix W (a symmetric, non-negative 101×101 matrix), the TasteRank score of variety i is the i-th component of the leading eigenvector $\mathbf{x}_1$, satisfying

$$W\mathbf{x}_1 = \lambda_1 \mathbf{x}_1,$$

where λ_1 is the largest eigenvalue. Three properties of this definition are essential to understanding TasteRank.

First, TasteRank is an implicit, system-level definition. There is no closed-form formula $\mathrm{TR}(i) = f(\cdots)$ that can be evaluated for a single variety in isolation. The 101 scores are defined by a system of 101 coupled equations that must be solved simultaneously — every variety's score depends on every other variety's score. This is exactly analogous to Google's PageRank, where no individual web page has an independent rank formula. A variety's TasteRank is not a local property of its profile or its neighbours — it is a global property of its position within the entire network.

Second, only the largest eigenvalue matters. When influence is propagated iteratively through the network ($\mathbf{z}^t = W^t \mathbf{z}^0$), the eigendecomposition shows that all contributions from $\lambda_2, \lambda_3, \ldots$ decay exponentially as $(\lambda_k/\lambda_1)^t \to 0$, leaving only $\mathbf{x}_1$ as the steady-state distribution. The spectral gap $|\lambda_2/\lambda_1| \approx 0.72$ for this graph ensures rapid convergence (35–45 iterations) and reflects a clear core–periphery structure.

Third, the logic is recursive. Written component-wise, the eigenvector equation becomes

$$x_i = \frac{1}{\lambda_1} \sum_j W_{ij} \, x_j.$$

A variety's score is proportional to the similarity-weighted sum of its neighbours' scores, which depend on their neighbours' scores, and so on to infinite depth. The eigenvector is the unique self-consistent solution to this infinite recursion. Sagrantino (rank 1) is connected to five other top-10 varieties, each connected to further high-centrality reds — recursive amplification through a dense, mutually reinforcing cluster. Riesling (rank 85) is connected to peripheral varieties whose neighbours are also low-centrality — recursive dampening through sparse, isolated connections. In TasteRank terms, distinctiveness and centrality are inversely related.

The full mathematical treatment — eigendecomposition proof, spectral gap analysis, power iteration algorithm, and the role of higher eigenvalues in spectral embedding — is provided in the *Technical Appendix* (Sections A.5.1–A.5.6).

3.4 PageRank as Structural Counterpoint

PageRank modifies eigenvector centrality by introducing a damping factor $\alpha = 0.85$ that models a random taster who follows similarity edges 85% of the time and teleports to a random variety 15% of the time. This prevents centrality from concentrating exclusively in the densest cluster and creates a diagnostic contrast: PageRank rewards bridge varieties connecting different communities, while eigenvector centrality rewards varieties deep within the densest core. The Spearman rank correlation between the two measures is $\rho \approx 0.92$, confirming robust agreement. The most informative divergence is Sangiovese (TasteRank rank 40, PageRank ~ 8) — a bridge spanning Communities C0 and C2. Full derivation in *Technical Appendix* A.6.

3.5 Community Detection

The Clauset–Newman–Moore greedy modularity algorithm partitions the graph into groups more densely connected internally than expected under a random null model, maximising

$$Q = \frac{1}{2m} \sum_{i,j} \Big[W_{ij} - \frac{s_i s_j}{2m} \Big] \, \delta(c_i, c_j).$$

The algorithm detects six communities with $Q \approx 0.41$, indicating strong structure. The number of communities is consistent with the eigenvalue spectrum of the modularity matrix, where the first six eigenvalues are clearly separated from the bulk. Full details in *Technical Appendix* A.7.

4. Results

4.1 TasteRank Rankings (Top 20)

The top 20 varieties by TasteRank are overwhelmingly full-bodied reds from Community C0, the Mediterranean/Southern Italian cluster. Sagrantino ranks first with a TasteRank of 0.3020 — its extreme profile (maximum colour depth, maximum tannin, maximum body) makes it the prototypical archetype of the densest cluster.

Rank	Variety	Type	Community	TasteRank
1	Sagrantino	Red	C0	0.3020
2	Nero d'Avola	Red	C0	0.2849
3	Lagrein	Red	C0	0.2748
4	Negroamaro	Red	C0	0.2380
5	Plavac Mali	Red	C0	0.2251
6	Montepulciano	Red	C0	0.2108
7	Petit Verdot	Red	C0	0.2093
8	Monastrell	Red	C0	0.2060
9	Petite Sirah	Red	C0	0.1768
10	Pinotage	Red	C0	0.1710
11	Malbec	Red	C0	0.1697
12	Tannat	Red	C0	0.1676
13	Garnacha Tintorera	Red	C0	0.1670
14	Dolcetto	Red	C0	0.1659
15	Mourvèdre	Red	C0	0.1565
16	Zinfandel	Red	C0	0.1498
17	Primitivo	Red	C0	0.1498
18	Merlot	Red	C0	0.1333
19	Carignan	Red	C0	0.1283
20	Limniona	Red	C2	0.1223

4.2 Community Structure

The modularity optimisation detects six communities. The most structurally significant finding is Community C1, which bridges the red–white divide: its four red members (Schiava, Poulsard, Frappato, Kadarka) are ultra-light reds with minimal tannin, positioning them closer to aromatic whites than to their nominal red peers.

ID	Character	Size	R / W	Hub Varieties
C0	Full-bodied Mediterranean Reds	31	31 / 0	Sagrantino, Nero d'Avola, Lagrein, Negroamaro
C1	Light / Aromatic Cross-boundary	21	4 / 17	Assyrtiko, Falanghina, Albariño, Riesling
C2	Mid-weight Structured Reds	18	18 / 0	Limniona, St. Laurent, Zweigelt, Blaufränkisch
C3	Rich Textural Whites	13	0 / 13	Marsanne, Fiano, Godello, Pinot Gris
C4	Mineral / Crisp Whites	10	0 / 10	Greco, Friulano, Verdicchio, Verdejo
C5	Lean Neutral & Aromatic Whites	8	0 / 8	Gewürztraminer, Petit Manseng, Malvasia

4.3 Key Structural Findings

Centrality concentration in Southern Italian/Mediterranean reds. The top 9 varieties all belong to Community C0. These are full-bodied, high-tannin, high-colour reds with profiles that cluster tightly in the thirteen-dimensional space. Sagrantino, Nero d'Avola, and Lagrein form the innermost core.

Noble varieties rank lower than expected. Cabernet Sauvignon ranks 30th, Pinot Noir 47th, Nebbiolo 42nd. These varieties have distinctive profiles that set them apart from the dense centre — high complexity and finish, but with unique trait combinations (e.g., Pinot Noir's low colour and tannin, Nebbiolo's extreme acidity and tannin with moderate colour). TasteRank measures centrality, not quality.

The red–white boundary dissolves in Community C1. Four light reds (Schiava, Poulsard, Frappato, Kadarka) cluster with aromatic whites. The cosine similarity between Frappato and Loureiro exceeds 0.98 — higher than Frappato's similarity to most other reds. This confirms algorithmically what sommeliers observe experientially.

White wine centrality peaks with medium-bodied textural varieties. Among whites, Marsanne (rank 54) and Fiano (rank 57) lead. These medium-bodied, moderately complex whites occupy the centre of the white-grape subspace. The most distinctive whites — Riesling (rank 85), Sauvignon Blanc (rank 90) — rank near the bottom, precisely because their extreme acidity and aromatic profiles set them apart.

5. Robustness Checks

5.1 TasteRank vs. PageRank: Empirical Comparison

As described in Section 3.4, PageRank provides a structural counterpoint to eigenvector centrality by introducing a teleportation mechanism that prevents extreme concentration. The Spearman rank correlation between TasteRank and PageRank across all 101 varieties is $\rho \approx 0.92$ — strong enough to confirm that the centrality structure is not an artefact of the choice of spectral measure, but with enough divergence to be informative.

The divergence pattern is systematic: varieties in the dense core of Community C0 have TasteRank > PageRank (recursive amplification within the cluster inflates their eigenvector centrality), while bridge varieties connecting multiple communities have PageRank > TasteRank (the random taster passes through them frequently when transitioning between clusters). Sangiovese is the most dramatic case: rank 40 by TasteRank but approximately rank 8 by PageRank, reflecting its 7 edges spanning Communities C0 and C2. The magnitude of the gap $|\text{TasteRank rank} - \text{PageRank rank}|$ thus serves as a diagnostic for structural role: large positive gaps identify core members, large negative gaps identify bridges.

5.2 Sensitivity to k

Varying the neighbourhood parameter k from 3 to 7 produces rank correlations of $\rho > 0.94$ for the top 30 varieties. The community structure is stable at $k = 4$ through $k = 7$ (six communities detected in each case). At $k = 3$, the graph fragments into seven communities as some white-grape components disconnect; at $k = 8+$, the graph becomes too dense and community resolution degrades. The choice of $k = 5$ represents a principled balance between sparsity and connectedness.

5.3 Spectral Gap Stability

The spectral gap $|\lambda_2/\lambda_1| \approx 0.72$ is stable across $k = 4{-}7$, confirming that the core–periphery structure is intrinsic to the data rather than an artefact of graph construction. A spectral gap this large indicates a single clearly dominant mode of centrality: the Mediterranean red cluster forms a robust gravitational core that persists regardless of parameter choices. The gap also ensures rapid convergence of power iteration (35–45 iterations at $\varepsilon = 10^{-6}$), making the computation numerically stable and reproducible.

6. Conclusion

6.1 Key Findings

TasteRank provides a principled, reproducible framework for understanding the structural organisation of the wine grape universe through the lens of sensory similarity. Three key findings emerge from the analysis.

First, centrality and quality are orthogonal. The varieties with the highest TasteRank — Sagrantino, Nero d'Avola, Lagrein — are not the world's most celebrated grapes. They are the most typical: their sensory profiles sit at the gravitational centre of the red-wine subspace, surrounded by many similar varieties. The noble varieties (Pinot Noir, Nebbiolo, Riesling) rank low precisely because their distinctive profiles push them to the periphery. In the TasteRank framework, distinctiveness and centrality are inversely related.

Second, the red–white boundary is permeable. Community C1 demonstrates that at the structural level, ultra-light reds (Schiava, Frappato, Poulsard) and aromatic whites (Loureiro, Müller-Thurgau, Welschriesling) share more sensory DNA than either group shares with its nominal colour peers. The tasting universe does not cleave neatly into red and white hemispheres — it contains a liminal zone where colour is secondary to structure.

Third, the framework is extensible. The same architecture can incorporate weighted expert similarity, regional terroir dimensions, or empirical tasting data from structured experiments. The TasteRank Explorer visualisation makes the network navigable and explorable, transforming an abstract mathematical structure into an interactive tool for wine education and discovery.

6.2 Limitations and Future Work

Three limitations bound the present results. First, the 13 sensory dimensions and the 0–5 scoring scale reflect modelling choices; an alternative dimensional framework (e.g., one that splits acidity into sensory and analytical components, or one that adds dimensions for fermentation-derived character) might yield different rankings at the margin. Second, the canonical-typicity scoring captures the centre of each variety's range but not its plasticity; varieties with wide stylistic ranges (Pinot Noir, Chardonnay) are summarised by a single profile that no individual wine fully embodies. Third, the 101-variety selection emphasises canonically recognised European and emerging regional varieties; further extension to a broader sample (Caucasian indigenous varieties, hybrid varieties, wider New World coverage) might surface additional cluster structure.

Future extensions could include: a vintage- or region-conditional layer that adjusts the canonical profile based on growing context; a separate aroma-compound-based encoding (terpenes, methoxypyrazines, thiols) to validate the structured-tasting dimensions against analytical chemistry; and integration with the Region Affinities and Region Resonances tools to allow joint queries across grape sensory, regional terroir, and cultural identity dimensions.

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